USB 2.0 + ULPI Hardware Connection Table

FPGA to ULPI PHY Connections

Clock and Reset Signals

FPGA Signal	FPGA Pin	ULPI PHY Signal	PHY Pin	Direction	Description
clk_50MHz	PIN_Y2	-	-	Input	50MHz system clock input
ulpi_clk_out	PIN_AB12	ULPI_CLK	3	Output	60MHz ULPI clock to PHY
ulpi_reset	PIN_AC12	RESETB	4	Output	PHY reset (active low)
reset_n	PIN_M23	-	-	Input	System reset button

ULPI Data Interface

FPGA Signal	FPGA Pin	ULPI PHY Signal	PHY Pin	Direction	Description
ulpi_data[0]	PIN_AE14	DATA0	13	Bidirectional	ULPI data bit 0
ulpi_data[1]	PIN_AF14	DATA1	14	Bidirectional	ULPI data bit 1
ulpi_data[2]	PIN_AG13	DATA2	15	Bidirectional	ULPI data bit 2
ulpi_data[3]	PIN_AH13	DATA3	16	Bidirectional	ULPI data bit 3
ulpi_data[4]	PIN_AE13	DATA4	17	Bidirectional	ULPI data bit 4
ulpi_data[5]	PIN_AF13	DATA5	18	Bidirectional	ULPI data bit 5
ulpi_data[6]	PIN_AG12	DATA6	19	Bidirectional	ULPI data bit 6
ulpi_data[7]	PIN_AH12	DATA7	20	Bidirectional	ULPI data bit 7

ULPI Control Signals

FPGA Signal	FPGA Pin	ULPI PHY Signal	PHY Pin	Direction	Description
ulpi_dir	PIN_AF12	DIR	11	Input	Direction control from PHY
ulpi_nxt	PIN_AG11	NXT	12	Input	Next data from PHY
ulpi_stp	PIN_AH11	STP	10	Output	Stop signal to PHY

User Interface Signals

FPGA Signal	FPGA Pin	External Connection	Description
read_button	PIN_M21	Push Button SW0	Manual read trigger
lba_manual_in[31:0]	PIN_G26:PIN_N25	DIP Switches	Manual LBA input
num_blocks_manual_in[15:0]	PIN_N26:PIN_P25	DIP Switches	Block count input
debug_info[31:0]	PIN_AE22:PIN_AF25	LEDs/7-Segment	Debug output display

Power Supply Connections

ULPI PHY Power Requirements

Supply	Voltage	Current	FPGA Pin	PHY Pin	Description
VDD33	3.3V	50mA	3V3 Rail	1, 2, 5, 6	Digital I/O supply
VDD18	1.8V	100mA	1V8 Rail	7, 8	Core supply
AVDD33	3.3V	30mA	3V3_ANALOG	21, 22	Analog supply
GND	0V	-	GND	9, 23, 24	Ground connections

USB Connector (Type-A Female)

USB Pin	Signal	ULPI PHY Pin	Description
1	VBUS	25	USB power (5V)
2	D-	26	USB data minus
3	D+	27	USB data plus
4	GND	28	USB ground
Shell	Shield	29	Cable shield

Additional Components Required

Crystal Oscillator (if not using FPGA PLL)

- **Component**: 60MHz crystal oscillator
- **FPGA Connection**: Connect to dedicated clock input pin
- **Load Capacitors**: 18pF to ground on each crystal pin

Bypass Capacitors

- **ULPI PHY**: 0.1μF ceramic + 10μF tantalum near each power pin
- **FPGA**: Follow manufacturer's power supply decoupling guidelines

Pull-up/Pull-down Resistors

Signal	Resistor	Value	Purpose
reset_n	Pull-up	10kΩ	Ensure defined reset state
ULPI_DIR	Pull-down	10kΩ	Default to output mode
ULPI_NXT	Pull-down	10kΩ	Default inactive state

PCB Layout Considerations

High-Speed Routing

- **ULPI_CLK**: Route as differential pair or controlled impedance (90Ω)
- **ULPI_DATA[7:0]**: Match trace lengths within ±0.1 inch
- **Ground Planes**: Solid ground plane under ULPI signals
- **Via Stitching**: Connect ground planes every 200 mils

EMI Considerations

- **Ferrite Beads**: On VBUS line (600Ω @ 100MHz)
- **ESD Protection**: TVS diodes on USB D+/D- lines
- **Shielding**: Ground pour around high-speed signals

Pin Assignment Notes

FPGA-Specific (Altera/Intel)

- Pins shown are for Cyclone IV EP4CE115 (DE2-115 board)
- For other FPGAs, adjust pin assignments accordingly
- Use FPGA vendor tools for optimal pin placement

ULPI PHY IC Examples

- **USB3300**: Microchip/SMSC USB 2.0 ULPI Transceiver
- **USB3320C**: Microchip Hi-Speed USB 2.0 ULPI PHY
- **ISP1504**: NXP USB 2.0 Hi-Speed ULPI Transceiver

Timing Constraints (SDC File)

```tcl
```

```
# ULPI Clock constraint
create_clock -name ulpi_clk -period 16.667 [get_ports ulpi_clk_out]

# Input delays for ULPI signals
set_input_delay -clock ulpi_clk -max 3.0 [get_ports {ulpi_data[*] ulpi_dir ulpi_nxt}]
set_input_delay -clock ulpi_clk -min 1.0 [get_ports {ulpi_data[*] ulpi_dir ulpi_nxt}]

# Output delays for ULPI signals
set_output_delay -clock ulpi_clk -max 2.0 [get_ports {ulpi_data[*] ulpi_stp}]
set_output_delay -clock ulpi_clk -min -1.0 [get_ports {ulpi_data[*] ulpi_stp}]
```

Testing and Debug

Signal Integrity Verification

- **Logic Analyzer**: Monitor ULPI protocol transactions
- **Oscilloscope**: Verify clock quality and timing margins
- **USB Protocol Analyzer**: Capture USB transactions end-to-end

Common Debug Points

- **debug_info[31:0]**: Connect to LEDs or 7-segment displays
- **ULPI Register Reads**: Implement register read capability
- **State Machine Monitoring**: Export FSM states to test points